

in several positional parameters in the expansion.

- `[me@linuxbox ~]$ foo="This string is long."`
- `[me@linuxbox ~]$ echo "$foo" is ${#foo} characters long."`
- `'This string is long.' is 20 characters long.`
`${parameter:offset}`
`${parameter:offset:length}`

Using this expansion strings offset characters to a specified length.

- `[me@linuxbox ~]$ foo="This string is long."`
- `[me@linuxbox ~]$ echo ${foo:5}`
- `the string is long.`

If there are negative values for offset, then it starts from the end of the string, not from the beginning.

- `[me@linuxbox ~]$ foo="This string is long."`
- `[me@linuxbox ~]$ echo ${foo: -5}`
- `long.`
- `[me@linuxbox ~]$ echo ${foo: -5:2}`
- `lo`
- `${parameter#pattern}`
- `${parameter##pattern}`

This expansion is used to remove the leading portion of the string in the parameter. With `#` you remove the shortest match while for the `##` it removes the longest match. Here is a quick demonstration of this fact.

- `[me@linuxbox ~]$ foo=file.txt.zip`
- `[me@linuxbox ~]$ echo ${foo#*.}`
- `txt.zip`
- `[me@linuxbox ~]$ echo ${foo##*.}`
- `zip`
- `${parameter%pattern}`
- `${parameter%%pattern}`

This expansion (%) works by removing the texts from the end of the string characters, not from the beginning.

- `[me@linuxbox ~]$ foo=file.txt.zip`
- `[me@linuxbox ~]$ echo ${foo%.*}`
- `file.txt`
- `[me@linuxbox ~]$ echo ${foo%.*}`
- `file`
- `${parameter/pattern/string}`
- `${parameter//pattern/string}`